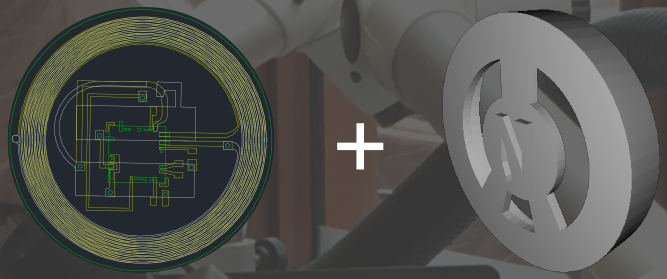
UV Sense



I joined the UV Sense team, which is part of the Rogers Research Group, in April 2018. The team was on a tight deadline to produce 300 devices that monitor UV exposure and sync to smartphones for L'Oréal (the sponsor). I was involved in every step of the process—from designing circuit boards, electroplating copper sheets, and soldering microscopic capacitors, MOSFETS, and UV sensors to devices to programming the devices. Although we created a device that is the smallest of its kind, the Rogers Research Lab wants to develop wearable devices that are practically invisible. Thus, last summer, they granted me a summer fellowship to explore how to redesign the UV sensor's antenna (which is its biggest component) so it could be smaller while maintaining functionality. Creating the antenna included much trial and error; however, the experience was hands-on and it strengthened my computer aided design skills, as well as my critical thinking and analytical skills. For example, when assembling the device to connect with the evaluation board or the phone, there was an issue with the connection. At first, I assumed the device was too small; however, after testing, it turned out that one of the capacitor on the pin was broken. Figuring out that issue took much troubleshooting and testing, but I ultimately built a useful and innovative product! Based on my work, another research team in the group has expressed interest in using my design in their pollution-detector device. In return, I will be credited in their published research.

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